GEOMETRICALLY ORDERED DYNAMICS
FOLIO F



The Anti-Drift Cognitive Control Loop

A Deterministic Coherence Gate for Sovereign Reasoning

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RESULTS STATED IN TIER — LIMITS FIRST

CONVERGENCE -- THE OBSERVER (ARI / sovereign cognition)

This folio converges on **the Observer** — sovereign cognition as an engineering discipline: keeping an autonomous cognitive loop inside its coherence band by drift control rather than by hope. Its unique angle is the **deterministic coherence gate**: a zero-external-dependency scoring gate (ADCCL), a cryptographic capability token (ARCHON), and a machine-checked control law ($u \geq \gamma$). Sibling papers approach the same claim from other angles: [TIER: CONJECTURE] the dipole folio (IT-FIELD) reads the Observer as ternary decision logic, the network folio (IT-NEURO) reads it as a biological substrate, and the starpolygon folio (IT-OMEGA) reads it as a layered architecture. Limits first: the 18/19 benchmark figure is not a generalization estimate; the v5 test and proof counts are carried as author-reported without deposited evidence; the geometric reading of the score is an interpretive scaffold until embedding cosine replaces the heuristic.

SUPERSESSION NOTE

This paper supersedes Zenodo record 21450415 ("Anti-Drift Cognitive Control Loop: A Geometric Coherence Gate for Sovereign Reasoning," 2026-07-20), retaining its record as the canonical folio. Record 21399702 is byte-identical to 21450415 and is absorbed, not republished. Record 21711417 (ADCCL v5, 2026-07-30) is absorbed for its one load-bearing mechanism (the ARCHON capability token); its asserted test and proof counts are not republished as verified. The retitle drops "Hallucination-Free," which appears in the old record's metadata but in none of its evidence. Corrections: the τ/θ threshold confusion is resolved by naming the two quantities distinctly (§2); the penalty weights' empirical basis is disclosed (§1); the v5 "89/89 tests" and "zero-sorry" claims are carried as author-reported, not verified (§4).

1 The gate

Unconstrained recursive execution in LLM agents produces structural degradation over extended task horizons: stub output, capability refusals, circular reasoning, hedging inflation, and semantic topic drift. The Anti-Drift Cognitive Control Loop (ADCCL) is a deterministic, zero-external-dependency heuristic scoring gate, implemented in Rust, that intercepts degraded output at the pipeline boundary before it corrupts persistent state. Every response passes through verify(response, task) before reaching persistent state; the function returns a **VerificationResult** carrying a verdict, a composite score, and a flag vector. No network calls, no secondary model invocations, no GPU.

The base score initializes at 1.0 and is decremented by fixed penalty weights per detected signal; penalties are additive and independent, and the final score is clamped to $[0, 1]$. [TIER: EMPIRICAL]

Signal	Detection	Penalty
STUB_MARKERS_DETECTED	regex: TODO, FIXME, XXX, STUB, [INSERT…]	−0.60
RESPONSE_TOO_SHORT	len < 40 bytes	−0.35
CAPABILITY_REFUSAL	regex: “as an AI”, “I can’t help”	−0.25
NO_TASK_WORD_OVERLAP	zero shared tokens (len > 3)	−0.30
CIRCULAR_RESPONSE	>70% task-token echo, length ratio < 1.5	−0.20
LOW_BIGRAM_COHERENCE	zero shared content words (len > 4)	−0.20
INCOMPLETE_SENTENCES	len > 50, no terminal punctuation	−0.10
LOW_INFORMATION_DENSITY	stopword ratio > 65%	−0.15
EXCESSIVE_HEDGING	hedge rate > 3 per 200 words	−0.15
HARM_POTENTIAL	regex: dehumanizing or violent language	−0.30

STUB_MARKERS_DETECTED carries a secondary hard veto: a flagged stub is rejected regardless of composite score. The crate declares two runtime dependencies (regex, serde) and compiles to both rlib and cdylib.

Disclosure. The penalty weights were set by engineering judgment during development on a 19-case synthetic suite constructed to exercise known failure modes. They were *not* calibrated against a labeled dataset, and the reported 18/19 correct-classification figure is *not* a generalization estimate: it measures behavior on inputs designed to trigger or avoid each signal. True false-positive and false-negative rates on production traffic are unknown. The **chiral_resonance** bonus (keyword/symbol density rewarding formal reasoning) evaluated to 0.0 on all 19 cases — none of the synthetic inputs contained formal symbols — so the reported behavior is dominated by the penalty model under ordinary prose conditions.

2 Two thresholds, named distinctly

The predecessor folio carried two numbers under overlapping names, and this successor fixes the confusion by naming them as different quantities.

The operational gate $\tau = 0.9539$. The rejection threshold used by verify(). Empirically observed during Chyren development as the stability boundary below which autonomous reasoning loops exhibited consistent semantic collapse; set as a constructor parameter, bounded below at 0.9539 regardless of session-progressive calibration. Portability to other pipelines is an open question; the claim that 0.9539 is a universal stability boundary is *not* supported by the current evidence.

The alignment floor $\theta = 1/\sqrt{2} \approx 0.7071$. The chiral-alignment gate below which a response is not merely degraded but misaligned. Earlier records asserted 0.7; the successor consolidates on the *derived* form. The derivation is the open-quantum-systems leg absorbed from the companion paper *Sovereign Cognition*: under a controlled Lindblad (GKSL) master equation with control rate u and environmental drift rate γ, the steady-state alignment satisfies $\chi_\infty = \sqrt{u/(u+\gamma)}$, hence $\chi_\infty \geq 1/\sqrt{2}$ iff $u \geq \gamma$. Sovereignty is dynamically enforceable iff the control rate meets the drift rate. [TIER: THEOREM]

The band is therefore $[\theta, \tau] = [1/\sqrt{2},\ 0.9539]$: the floor is derived, the ceiling is empirical. The old figure’s $[0.7, 0.9539]$ band is superseded by this naming.

3 What is proved

The boundary equivalence. In Lean 4, with Mathlib, the threshold is placed on a rigorous footing as an exact statement about the unit interval:

`chiral_stability_boundary` : for nonzero Ψ, Φ in an inner product space,

$$\text{chiral_invariant}(\Psi, \Phi) \geq 0.953939 \iff \text{angle}(\Psi, \Phi) \leq \arccos(0.953939),$$

where chiral_invariant = cos(angle). `[TIER: THEOREM]` The proof contains no `sorry` and invokes no bespoke axiom; the compiler's dependency report is `[propext, Classical.choice, Quot.sound]` — Lean's standard trust base and nothing else. What the proof does *not* establish: that the ADCCL heuristic score is geometrically a cosine of embedding vectors. The geometric framing is an interpretive scaffold; a future embedding-cosine implementation would inherit the theorem directly.

The published correction. Version 1 of the folio stated the boundary with a decimal angle (0.3040), claimed both directions "verified numerically," and discharged the statement by axiom in the Lean source. None of that was so: $\arccos(0.953939) = 0.3046933\ldots$, so the decimal claim was false, and no finite decimal repairs it. Version 2 states the bound exactly as $\arccos(\chi_s)$ and proves it outright. This correction notice is carried forward as the record's integrity artifact: the false claim was admitted in print, not silently edited.

The control law. Absorbed from *Sovereign Cognition* (Drive predecessor, 2026-09-28): the anti-drift dissipator $U[\rho, \ell_t] = \sum_i u_i(\ell_t)\, D[C_i](\rho)$ with $u_i \geq 0$ makes the full generator GKSL, hence completely positive and trace-preserving (Theorem 1; the trace-preservation core is Lean-formalized, axiom-clean; full Choi verification is future work). Theorem 2 is the $u \geq \gamma$ enforcement threshold above. The Sovereignty Score components $\Omega = \alpha\Omega_S + \beta\Omega_B + \delta\Omega_P$ are given explicit definitions (entropy suppression, boundary resonance, Berry phase) and shown mutually consistent — Ω_S vanishes at the same $u = \gamma$ point where the χ-gate fails. Carried limits: the weights α, β, δ are modeling choices, not derived; Ω_P is defined up to Berry-phase sign convention; the cognitive interpretation is untested as an agent mechanism.

4 The capability-token layer

Absorbed from ADCCL v5 (21711417), the one load-bearing mechanism beyond the folio: the ARCHON zero-trust capability verifier. Every high-privilege execution request must present a capability token signed via HMAC-SHA256,

$$\sigma = \text{HMAC}{-}\text{SHA256}(K_{\text{secret}},\ \text{grant_id} \,\|\, \text{capability} \,\|\, \text{expiry}), \tag{1}$$

under the three-way conjunction $\text{Verdict} = \text{Gate}_{\text{AEGIS}} \wedge \text{Gate}_{\text{Solomonic}} \wedge \text{Gate}_{\text{ARCHON}}$, with 0 = Suspend, -1 = Abort. This is a real cryptographic governance primitive: policy enforcement embedded in the state machine execution path rather than applied as an external filter. `[TIER: THEOREM]` (specification-level: the construction is exhibited; its deployment is reported, not independently audited here.)

Honest block (v5 claims not carried as verified). Record 21711417 asserts "89/89 unit tests passing" and "Lean 4 proofs with zero-sorry placeholders" with no test logs, sources, or proof artifacts deposited in-record. These figures are carried here as *author-reported*, not verified. "Double Helix Phase Locking" restates the two legacy constants ($\kappa \leq 0.953926$, $\theta \geq 0.700000$) as helical strands with no dynamics; "Solomonic gates" and "Bloch sphere contradiction screening" are undefined in-record. None of these terms is used as an established result in this successor.

5 Limits

The evaluation harness documented two incorrect verdicts, and both define the guarantee boundary. A regex false positive on ordinary technical vocabulary was diagnosed and fixed via a two-pattern post-filter (precision failure, resolved). A paraphrase-constructed circular response passed with chiral_invariant = 1.0 (recall failure): the detector operates on lexical token overlap, not semantic similarity, and this failure is *not patchable within the lexical framework* — it requires the embedding-cosine architecture of §3's forward-compatibility note. Semantic drift without lexical degradation is invisible to the current system. The threshold $\tau = 0.9539$ has not been validated against external agent systems or alternative providers. The Drive companion *The_Observerse* (E8 branching, on the visible-sector factor of the E8×E8 canon, for the 14-dimensional observerse) was absorb-checked and carries no drift-control content; its material belongs to the Shadow-ladder lane and is left out of this folio with that reason recorded.

References

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